

The Utility Problem

Dynamic Line Rating (DLR) is a technique used in distribution grids to optimize the use of existing electrical transmission and distribution lines (power lines). It provides real-time information about the actual operating conditions of a power line, such as ambient temperature and wind speed, and uses this information to determine the safe and efficient operating capacity of the line.

In traditional systems, power lines are rated based on conservative and fixed criteria, such as maximum ambient temperature and wind speed. This often results in significant unused capacity, especially during periods of low ambient temperature and higher wind speed. Power lines can carry more current in the presence of moderate to high winds because the wind carries away the heat being dissipated by the power lines. On the other hand, in extreme heat and windless conditions, this static capacity can be exceeded. In very warm weather, when temperatures exceed 35 °C, the dynamic load capacity may fall below the static load capacity. In this case, the line must be loaded only according to the dynamic capacity.

With DLR, the capacity of a power line is continuously monitored and updated in real-time based on the actual operating conditions. This allows for the safe and efficient utilization of existing capacity, reducing the need for costly upgrades to the grid. DLR systems can increase the power capacity of the existing grid anywhere from 10%-30% to more than 90% of the time ¹.

DLR can also help to improve the reliability and quality of service of a distribution grid, by preventing overloading and reducing the risk of power outages. Additionally, DLR can support the integration of renewable energy sources, such as wind and solar, by allowing for the efficient use of existing transmission lines to carry the generated power to the customers. Wind generation is, obviously, located in windy areas and can benefit considerably from dynamic line ratings. A slight increase of 2 feet per second in wind speed can increase the ampacity of a transmission line by 15%². It can also be used, for example, when repairing part of the grid, when it is necessary to reconnect part of the customers to another line. If this line with new customers is at the edge of maximum load, it is possible, for example, in winter when lower temperatures are expected, to use this unused capacity caused by lower ambient temperature (due to better heat transfer from the conductor). In general, the methodology can also be used for grid planning - when it is colder, customers heat more and more power needs to be delivered to those customers, but it is possible to use the unused capacity caused by the lower ambient temperature.

In summary, the DLR is a valuable technology that can help to improve the efficiency, reliability, and sustainability of a distribution grid. For this, it is important to have a system in place that allows the calculation of DLR.

¹ Sandy K. Aivaliotis, Dynamic line ratings for optimal and reliable power flow. Accessed on June 15, 2014. Retrieved from <https://www.ferc.gov/EventCalendar/Files/20100623162026-Aivaliotis,%20The%20Valley%20Group%206-24-10.pdf>

² Sandy K. Aivaliotis, Dynamic line ratings for optimal and reliable power flow. Accessed on June 15, 2014. Retrieved from <https://www.ferc.gov/EventCalendar/Files/20100623162026-Aivaliotis,%20The%20Valley%20Group%206-24-10.pdf>

The Data Solutions and Methods

This methodology is suitable for areas where grid capacity is close to maximum. Information about grid locations which are about to reach their limiting capacity can be obtained by a use case Feeder Capacity and Lateral Capacity Analysis (also located in the Awesense use cases library).

For these areas, we know the Ampacity for normal conditions, and we are able to calculate the Ampacity for worse and better ambient conditions:

- Ampacity calculated for normal ambient conditions - Static Ampacity.
- Ampacity calculated for low temperature and cloudy sky.
- Ampacity calculated for higher wind speeds.
- Ampacity calculated for ambient temperatures above 35 °C.

These 4 values are then entered into the Awesense Platform for each grid section and are therefore available for subsequent decision-making. Based on the weather forecast, the different ampacities can then be used for operational grid management but also for long-term grid planning. The remaining available ampacity can be then sent to grid management tools (like ADMS) for real-time decision making.

The Awesense Platform helps in integrating all the data that is necessary for this analysis. These are mainly GIS data with connectivity, but also time series data of the load on individual sections. In addition, the Awesense Platform offers handy weather API which can bring additional valuable information into the analysis.

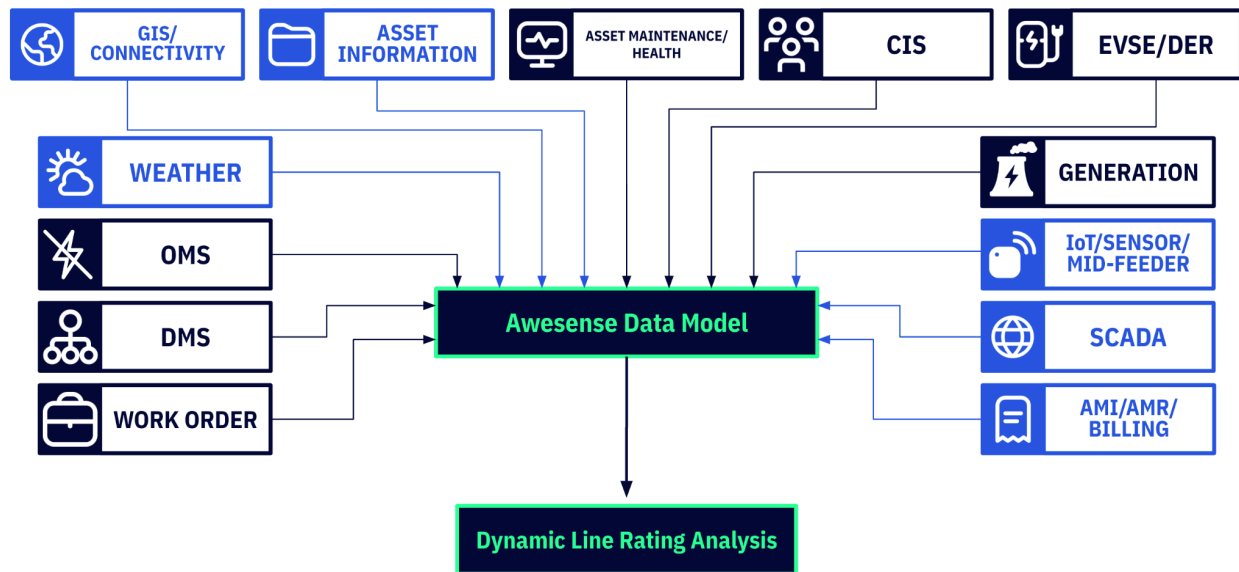
Dynamic Line Rating Analysis

[UC47-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.